

CIT-Sentinel

CIT-Sentinel — Final Round Pitch Script

Chennai Institute of Technology | SDG 4 — Quality Education

OPENING HOOK (45 seconds)

"Good morning everyone.

Let me start with a question.

How many of you remember a classmate who was brilliant in first year... and simply disappeared by third year?

They didn't fail overnight. They started missing one lab session. Then two. Their assignments came in late. Their LMS logins dropped to once a week. Their IAT scores slipped quietly. And nobody — not their teacher, not their mentor, not their HOD — caught it in time.

That student didn't fail because they weren't capable.

They failed because the system around them was blind.

We built CIT-Sentinel to give that system eyes."

THE PROBLEM (60 seconds)

"Let's talk about the real scale of this problem.

India loses thousands of engineering students to dropout every single year. Not because of lack of intelligence. But because of lack of early detection.

At any Anna University affiliated college — including Chennai Institute of Technology — a faculty member manages 60 to 90 students per class. A mentor handles an entire batch. An HOD oversees a full department. Nobody has the bandwidth to manually track every silent signal from every student before it becomes a failing grade or a detention order.

And here's what makes it worse — traditional systems are completely reactive. They flag a student AFTER attendance crosses the detention threshold. AFTER IAT scores are already submitted and failed. AFTER the student has emotionally checked out and stopped showing up entirely.

The problem statement given to us is precise — integrate attendance patterns, LMS activity, assessment performance, behavioral indicators, competition participation, and historical academic records to generate predictive risk scores and trigger interventions BEFORE performance decline becomes irreversible.

That word — BEFORE — is everything.

That is the entire mission of CIT-Sentinel."

OUR SOLUTION — THE BIG IDEA (60 seconds)

"CIT-Sentinel is not a dashboard. It is not an attendance tracker. It is not a marks management system.

It is a living, breathing academic intelligence platform that watches over every student in the institution simultaneously — processing six dimensions of behavioral data in real time — and surfaces the right information to the right person at the right moment in the academic calendar.

The core insight that drives our entire system is this:

Dropout is not an event. It is a process. And that process leaves traces — behavioral traces — weeks before any grade officially fails.

A student starts skipping lab sessions specifically. Not theory. Just lab. That is a signal.

A student's LMS logins are high but their IAT score is low. That is a different signal.

A student wins three hackathons but fails their model exam. That is yet another signal.

Each signal points to a different root cause. And each root cause demands a completely different intervention.

CIT-Sentinel reads all of these signals. Classifies them. And tells the right person exactly what to do — before it is too late."

THE SENTINEL SCORE (45 seconds)

"At the heart of our platform is the Sentinel Score — a composite 0 to 100 risk indicator computed for every student using five weighted data dimensions.

Attendance velocity carries 30 points — not just the current percentage, but how fast it is dropping.

IAT academic performance carries 25 points across both assessments.

LMS digital engagement carries 20 points — are they even logging in to study?

Model exam trajectory carries 15 points — are they building toward the end semester?

Competition and hackathon participation carries 10 points — are they technically active even if academically struggling?

A score above 75 — the student is safe.

Between 50 and 74 — early warning, monitoring required.

Below 50 — active intervention triggered immediately.

This single number gives every stakeholder in the institution — student, teacher, mentor, HOD, principal, chairman — an instant understanding of where every student stands. Right now. Not after exams. Not after results. Now."

PATTERN RECOGNITION ENGINE (75 seconds)

"But what truly separates CIT-Sentinel from every other academic monitoring system is our Pattern Recognition Engine.

We don't just say a student is at risk. We say WHY they are at risk. And we prescribe a specific intervention for that specific pattern.

We have identified seven distinct behavioral patterns.

The TECHNICAL STAR pattern — a student winning hackathons but failing theory. This student is not lazy. They are misaligned. They need a Theory-Bridge — not a warning letter. We pair them with a subject topper for structured weekly revision.

The CRAMMING PATTERN — high LMS activity but low IAT scores. This student is studying hard but not effectively. They are memorizing, not understanding. They

are at high risk of burnout and sudden dropout after exam failure. They need active recall strategies, not more reading material.

The LAB DRIFT pattern — attendance drops specifically on lab days while theory attendance stays stable. This signals practical disengagement — possibly a scheduling conflict, a fear of hands-on assessment, or early-stage burnout with practical work. Completely different root cause. Completely different fix.

The THEORY DRIFT pattern — the inverse. Strong lab presence, declining theory attendance. Often signals lecture-style learning incompatibility. Micro-learning video recommendations are triggered automatically.

The FULL DROPOUT pattern — multi-dimensional failure across attendance, marks, and digital engagement simultaneously. This triggers our most urgent protocol — immediate parent contact, HOD escalation, and emergency counseling referral.

The BURNOUT pattern — a sharp attendance velocity drop combined with multiple late submissions in a student who was previously performing well. Something has changed in their life. A counselor, not a marks reminder, is what they need.

Seven patterns. Seven interventions. Because one warning letter never saved anyone."

ROLE-BASED PLATFORM WALKTHROUGH (2 minutes)

"Now let me walk you through the platform itself.

CIT-Sentinel is built around the real institutional hierarchy of an engineering college. Six roles. Six completely different interfaces. Each person sees exactly what they need — nothing more, nothing less.

THE STUDENT logs in with their register number. The very first thing they see is their Sentinel Score — displayed as an animated circular ring, color-coded by risk level. Not a grade. Not a percentage. A single number that tells them: here is where you stand right now, here is why, and here is what you should do next.

Below the score, personalized insights in plain language — not technical jargon. Not 'Your attendance velocity has declined 12 percentage points.' Instead: 'You

need to attend 6 more classes this week to stay above the UGC minimum. Your lab attendance is strong but you're missing theory sessions.'

The student also has access to Sentinel Chat — a private AI counselor powered by Claude. The counselor already knows their academic context. But it speaks like a supportive senior — warm, non-judgmental, motivating. Not an alarm system. The conversation is completely private. Their mentor cannot see it. This is crucial — students need a safe space to be honest about their struggles before they can accept help.

THE SUBJECT TEACHER sees the entire class in a structured performance table — IAT 1, IAT 2, Model Exam, Attendance, and a Sentinel Score mini-bar for every student. One click on any row reveals that student's behavioral pattern, AI-generated recommendation, and quick-action buttons to refer them to their mentor or push concept video recommendations directly to their dashboard.

THE MENTOR has the most operationally powerful view. All 60 assigned mentees as interactive cards — showing attendance, IAT scores, LMS activity, drift pattern, and a one-line AI reasoning note explaining exactly why that student was flagged. The Peer-Bridge system automatically matches an at-risk student with the highest-performing compatible student in the same section — turning the student community into a support network. The AI Intervention Planner generates a complete 4-week action plan for each critical student — specific daily actions, check-in questions for each week, what NOT to say, and the exact opening line the mentor should use when approaching that student. The entire plan is presented as a checkable task list the mentor tracks week by week.

THE HOD sees the department command center. Retention rate, average Sentinel Score, active interventions, and critically — the Weekly Risk Evolution chart showing how the Critical, At-Risk, and Safe populations have moved across 8 weeks of the semester. The Sentinel Trigger Window at Week 3 is clearly marked. This chart answers the HOD's most important question: are we catching students early enough?

The Dropout Insights AI page goes further — it performs batch-level analysis of the four major dropout causes detected in this cohort — Attendance Collapse, Hidden Talent Loss, Academic Overload, and Disengagement — and delivers strategic prevention recommendations specific to Anna University's academic calendar and examination structure.

THE PRINCIPAL manages cross-departmental compliance and escalation. Urgent alerts surface students with unresolved critical interventions. The Compliance Tracker shows institution-wide adherence to UGC attendance regulations with detention-risk students immediately visible. No student approaching the 65% detention threshold goes unnoticed.

THE CHAIRMAN sees the full strategic picture. A Department Performance Radar Chart comparing all four departments across five dimensions simultaneously. An Industry Readiness quadrant plot dividing students into four categories — Industry Stars, Hidden Talent, Academic Focus, and Needs Support. And the SDG 4 Impact dashboard — which we will come to in a moment."

THE AI ENGINE — DROPOUT ORACLE (45 seconds)

"Every role in the platform also has access to our flagship AI feature — the Dropout Oracle.

Any staff member selects any student and runs a full dropout risk analysis. The system sends that student's complete behavioral profile — attendance velocity, LMS patterns, IAT trajectory, drift classification, competition history — to our AI engine, which returns a structured report in seconds.

The report contains: a dropout probability percentage. The top three root causes specific to that student. A week-by-week 4-week rescue plan with concrete daily actions. Common intervention mistakes to avoid. And a suggested opening line the mentor can use when approaching that student face to face.

This is not generic advice. This is contextual, data-grounded intelligence that understands the difference between a student who is disengaged and a student who is overwhelmed. And it prescribes accordingly.

For a mentor managing 60 students, this changes everything. Instead of spending 3 hours trying to understand why a student is struggling, they have a complete action plan in 30 seconds."

NOVELTY AND DIFFERENTIATION (45 seconds)

"So what makes CIT-Sentinel genuinely novel?

Every college has an attendance system. Every college has a marks portal. What no college has is a system that connects all of these data points, identifies the behavioral pattern behind the numbers, and prescribes a human-centered intervention — automatically — before the student even realizes they are at risk.

Our key differentiators are three:

First — Week 3 detection. Not Week 14. The intervention window in an Anna University semester is narrow. We catch the signal before the window closes.

Second — Pattern-specific intervention. Not one generic warning for all risk types. Seven patterns, seven strategies, each matched to the root cause.

Third — Role-appropriate intelligence. Every stakeholder sees exactly what they need to act, at the level of granularity appropriate to their role. A student doesn't need to see intervention logs. A chairman doesn't need to see individual assignment submission records. CIT-Sentinel shows the right data to the right person every time."

SDG 4 IMPACT (30 seconds)

"This project directly addresses UN Sustainable Development Goal 4 — Quality Education.

Our semester simulation shows that with Sentinel intervention activated at Week 3, projected dropout probability across a 16-week semester drops from 65% to under 12% for at-risk students.

Every student flagged early. Every intervention triggered on time. Every dropout that doesn't happen.

That is our SDG 4 contribution. Inclusive education. Quality outcomes. Data-driven equity. For every student — regardless of background, learning style, or how quietly they struggle."

TECHNOLOGY (20 seconds)

"CIT-Sentinel is built on React 18, TypeScript, Tailwind CSS, and Recharts for the frontend — deployed on Vercel for instant global access. AI features are powered by the Anthropic Claude API with role-specific system prompts. Authentication

ensures strict data isolation between all six roles. The platform is designed for local campus server deployment so all student data stays within the college network — fully privacy compliant."

CLOSING (30 seconds)

"We came into this hackathon with one question:

What if a college could know — really know — which student was about to fall through the cracks, three weeks before it happened?

CIT-Sentinel is our answer.

Not a system that punishes students for failing. A system that refuses to let them fail alone.

Built for Chennai Institute of Technology. Built for every student who deserved someone to notice sooner.

Thank you."

Total runtime: approximately 9 to 10 minutes at a confident, natural pace.

If you need a 5-minute version — cut the Role-Based Walkthrough to 45 seconds by saying:

"Six roles — student, teacher, mentor, HOD, principal, and chairman — each with a completely different purpose-built interface ensuring the right data reaches the right person. A student sees only their own personalized dashboard. A mentor gets AI-generated intervention plans for every critical mentee. A chairman sees institution-wide strategic intelligence."

Then jump straight to the AI Engine section.